

A NOTE ON NODALLY BOUND-PRESERVING FINITE ELEMENT METHODS FOR THE GINZBURG–LANDAU ENERGY

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ABSTRACT. test

1. INTRODUCTION

$$E(u, \mathbf{A}) := \frac{1}{2} \int_{\Omega} \left| \frac{i}{\kappa} \nabla u + \mathbf{A} u \right|^2 + \frac{1}{2} (1 - |u|^2)^2 + |\operatorname{curl} \mathbf{A} - \mathbf{h}_{\text{ext}}|^2 \, dx,$$

Here, we only study the reduced Ginzburg–Landau model with given $\mathbf{A}$:

$$(1.1) \quad E(u) := \frac{1}{2} \int_{\Omega} \left| \frac{i}{\kappa} \nabla u + \mathbf{A} u \right|^2 + \frac{1}{2} (1 - |u|^2)^2 \, dx,$$

since this

Some literature:

- [16] original physis paper
- Du et al : [10, 11], [13] and [12]
- so-called Abrikosov vortex lattice [1]
- structure of the vortex lattice, see [17–20]. There are no vortices for small values of κ , while larger values lead to more and more (very localized) vortices [2, 17].
- [9] reduced problem, Lagrange
- [5], reduced problem, LOD
- [6], pollution, Lagrange
- [14], pollution + LOD
- [7] full problem, smooth
- [8], full problem, non-convex domains
- time-dependent GLE with maximum bounds: [15]

2. GINZBURG–LANDAU PROBLEM AND DISCRETE MINIMIZERS

2.1. Preliminaries. For the mathematical framework, we require the following notation: For $2 \leq p \leq \infty$ we denote by $L^p = L^p(\Omega, \mathbb{C})$ the standard Lebesgue space of scalar complex-valued function on Ω equipped with the usual norm $\|\cdot\|_{L^p}$. For $p = 2$ the Hilbert space L^2 is considered as a *real* Hilbert space with the inner product

$$(v, w) = \operatorname{Re} \int_{\Omega} v \bar{w} \, dx$$

where $\bar{w}$ denotes the complex conjugate of w .

The first Fréchet derivative of the energy functional

$$E'(u) = \partial_u E(u) : H^1 \rightarrow \mathbb{R}$$

is given by

$$\langle \partial_u E(u, \mathbf{A}), v \rangle = \left(\frac{i}{\kappa} \nabla u + \mathbf{A} u, \frac{i}{\kappa} \nabla v + \mathbf{A} v \right) + (|u|^2 - 1)u, v, \quad v \in H^1(\Omega)$$

By the first-order optimality condition each minimizer u , if it exists, needs to be a critical point of E , i.e., $\langle E'(u)v \rangle = 0$ for all $v \in H^1$. This leads to the famous Ginzburg-Landau equation

$$\left(\frac{i}{\kappa} \nabla u + \mathbf{A}u, \frac{i}{\kappa} \nabla v + \mathbf{A}v \right) + ((1 - |u|^2)u, v) = 0$$

for all $v \in H^1(\Omega)$.

B: Something on gauge invariance here, and the fact that we need to incorporate everything in $(iu)^\perp$.

B: define H_{iu}^1

Lemma 2.1. *coercivity of E'' with $\rho_\kappa(u)$*

- Recall some results on a-priori estimates of minimizers, in particular $|u| \leq 1$ a.e.
- stuff on E'' in particular $\rho_\kappa(u)$
- ...

$$\begin{aligned} \langle \partial_u^2 E(u)[w, v] \rangle &= \operatorname{Re} \int_{\Omega} \left(\frac{i}{\kappa} \nabla w + \mathbf{A}w \right) \left(\frac{i}{\kappa} \nabla v + \mathbf{A}v \right)^* \\ &\quad + (|u|^2 - 1)wv^* + u^2 w^* v^* + |u|^2 wv^* \, dx \end{aligned}$$

Note the identity which implies

$$(2.1) \quad \langle \partial_u^2 E(u)[u, v] \rangle = \langle \partial_u E(u)v \rangle + \operatorname{Re} \int_{\Omega} 2|u|^2 uv^* \, dx$$

Important inner product

$$(2.2) \quad a_\kappa(w, v) = \operatorname{Re} \int_{\Omega} \left(\frac{i}{\kappa} \nabla w + \mathbf{A}w \right) \left(\frac{i}{\kappa} \nabla v + \mathbf{A}v \right)^* \, dx + \beta \operatorname{Re} \int_{\Omega} wv^* \, dx$$

with $\beta = 1 + \|\mathbf{A}\|_{L^\infty}^2$.

Lemma 2.2. **B:** *Cite the regularity results here*

3. MINIMIZATION UNDER MBP CONSTRAINT

In order, to have some more flexibility in the analysis, we do not set the constraint to 1, but also take some $\delta \geq 0$, and constrain to $1 + \delta$. We treat the $\delta = 0$ and $\delta > 0$ differently. Let $V_h \subset H^1(\Omega)$ being an N -dimensional finite element space with nodal Lagrange basis functions φ_i , nodes $\mathbb{A} = \{x_i \mid i = 1, \dots, N\}$, and degrees of freedom $\xi_i \in \mathbb{C}$ for $i = 1, \dots, N$. For any $v_h = \sum_{i=1}^N \xi_i \varphi_i \in V_h$ let us define the corrected component v_h^+ and the error component v_h^- via

$$\xi_i^+ = \begin{cases} \xi_i & \text{if } |\xi_i| \leq 1 + \delta \\ \xi_i \frac{1+\delta}{|\xi_i|} & \text{if } |\xi_i| > 1 + \delta, \end{cases} \quad v_h^+ = \sum_{i=1}^N \xi_i^+ \varphi_i, \quad v_h^- = \sum_{i=1}^N (\xi_i - \xi_i^+) \varphi_i.$$

Clearly, we have $v_h^+ = (v_h^+)^+$, and for $g(x) = |x|^2 - (1 + \delta)$ the convex set of positive components, where we also intersect with $(iu)^\perp$: $V_h^{iu} = V_h \cap (iu)^\perp$

$$\begin{aligned} V_h^+ &:= \{\varphi_h^+ \mid \varphi_h \in V_h\} = \{\varphi_h \in V_h \mid |\varphi_h| \leq 1 + \delta \text{ at all } x_i \in \mathbb{A}\} \\ (3.1) \quad &= \{\varphi_h \in V_h \mid g(\varphi_h) \leq 0 \text{ at all } x_i \in \mathbb{A}\} \\ V_h^{+,iu} &:= V_h^+ \cap (iu)^\perp \end{aligned}$$

is the space of admissible solutions if linear finite elements are considered.

We aim to find $E(v_h) \rightarrow \min$ in $V_h^{+,iu}$. The critical point criterion for (components of) minima in the interior of V_h is that the gradient $E'(v_h) = 0$ vanishes, while for critical points on the boundary, the gradients $E'(v_h)$ and $g'(v_h)$ need to be linearly dependent. That is, there is a $\lambda \in \mathbb{R}$, usually

called Lagrange multiplier, such that $E'(v_h) = \lambda g'(v_h)$. If $\lambda > 0$, the critical point cannot only be a local minimum, while $\lambda < 0$ excludes a local maximum.

We now want to study the minimization problem

$$(3.2) \quad E(u_h) = \min_{v_h \in V_h^+} E(v_h),$$

Following [9], we may extract a sequence of minimizers as $h \rightarrow 0$ such that $u_h \in V_h^{+,iu}$ and $u_h \rightarrow u$ in $H^1(\Omega)$. For the rest of this work, we fix this sequence.

For the error analysis, a crucial tool is the constrained Ritz projection R_h for the inner product $a_\kappa(\cdot, \cdot)$ defined in (2.2) via

$$(3.3) \quad R_h u = \operatorname{argmin}_{v_h \in V_h^+ \cap (iu)^\perp} \frac{1}{2} \|u - v_h\|_{a_\kappa}^2.$$

We note that R_h is well-defined by the strict convexity of the functional and the feasible set. However, the operator is not linear. Further, we obtain the boundedness

$$\|R_h u\|_{a_\kappa} \leq \|u\|_{a_\kappa} + \|u - R_h u\|_{a_\kappa} \leq 2\|u\|_{a_\kappa},$$

where we used that $0 \in V_h^{+,iu}$. In addition, we immediately observe by the definition of R_h

$$\|u - R_h u\|_a \leq \inf_{v_h \in V_h^+ \cap (iu)^\perp} \|u - v_h\|_{a_\kappa}.$$

With this, we can present our first main result for two at the moment independent minimizers.

Theorem 3.1. (a) Let $u \in H_\kappa^1$ be a minimizer of (1.1) and $u_h \in V_h^{+,iu}$ a minimizer of (3.2). Then, we have the estimate

$$\|u - u_h\|_{H_\kappa^1} \leq \|u - R_h u\|_{H_\kappa^1} + \rho_\kappa(u)(2 + \beta)\|u - R_h u\|_{L^2} + \rho_\kappa(u)(3\|u - u_h\|_{L^4}^2 + \|u - u_h\|_{L^6}^3)$$

(b) In addition, let u_h be a sequence that converges to u in $H^1(\Omega)$. Then, for h sufficiently small it holds

$$\|u - u_h\|_{H_\kappa^1} \leq \|u - R_h u\|_{H_\kappa^1} + \rho_\kappa(u)(2 + \beta)\|u - R_h u\|_{L^2}$$

with constants independent of κ and h .

This result is proved in Section 4. The essential tool used in the proof is the variational inequality satisfied by the Ritz projection. However, no approximation properties are used so far, and the above statement holds independently of these results.

3.1. Main results on the Ritz projection. In order to conclude an error bound from Theorem 3.1, we now provide estimate on the Ritz projection. First, we consider the approximation in H_κ^1 , and obtain the following estimate. We note that at this stage this bound is only valid for the minimizer itself.

Theorem 3.2. Let $u \in H^2$ be a minimizer of (1.1), then

$$\|u - R_h u\|_a \lesssim \kappa h$$

with a constant independent of κ and h .

The proof is presented in Section 5.

However, in order to obtain convergence of possibly optimal order, we need approximation of the Ritz projection also in L^2 . In this manuscript, we present two results. The first result treats the relaxed case $\delta > 0$, but can deal with dimension $d = 2, 3$. Note that we could also state this result for the one-dimensional case, but as it is included in the second result, we refrain from doing so.

Theorem 3.3. *Let $\delta > 0$. Then, for h sufficiently small, in particular such that for constants C_∞, C'_∞*

$$\begin{cases} C_\infty \kappa^2 |\log h| h < \delta, & d = 2 \\ C'_\infty \kappa^2 h^{1/2} < \delta, & d = 3 \end{cases} \text{ sufficiently small}$$

it holds

$$\|u - R_h u\|_{L^2} \lesssim \kappa h \|u - R_h u\|_{a_\kappa} \lesssim (\kappa h)^2$$

with a constant independent of κ and h .

The proof is presented in Section 6. The second result mimics the techniques of [3] for $\delta = 0$, but is only valid for $d = 1$.

Theorem 3.4. *Let $d = 1$ and $\delta \geq 0$, then*

$$\|u - R_h u\|_{L^2} \lesssim \kappa h^{1/2-\varepsilon} \|u - R_h u\|_a \lesssim \kappa^2 h^{3/2-\varepsilon}$$

with a constant independent of κ and h .

The proof is presented in Section 7. In these two scenarios we indeed obtain optimal order of vconvergence for the minimizer in H_κ^1 .

Corollary 3.5. *Assume either the setting of Theorem 3.3 or Theorem 3.4. In both cases, we obtain under the assumptions of Theorem 3.1*

$$\|u - u_h\|_{H_\kappa^1} \lesssim \kappa h$$

where the constant is independent of κ and h .

4. ERROR ANALYSIS

In this section, we follow the arguments from [9], but have to take care of the nonlinear Ritz projection. Therefore, we first derive the variational inequality for the Ritz projection, the critical point equation of the minimizers, and finally prove the first main result.

4.1. Variational inequality for the constrained Ritz projection. Recall the space $V_h^{+,iu}$ from (3.1) and we see that (3.3) clearly satisfies

$$R_h u = \operatorname{argmin}_{v_h \in V_h^+ \cap (iu)^\perp} J(v_h)$$

for $J(v_h) = \frac{1}{2} \|u - v_h\|_{a_\kappa}^2$. With this, we prove the following lemma.

Lemma 4.1. *Let $R_h u$ be the unique minimizer of (3.3). Then, we have*

$$a_\kappa(u, v_h - R_h u) \leq a_\kappa(R_h u, v_h - R_h u)$$

for all $v_h \in V_h^{+,iu}$.

B: Stampacchia's theorem? Below, we exploit this relation for $v_h = u_h$ in the form

$$(4.1) \quad a_\kappa(R_h u - u, R_h u - u_h) \leq 0$$

since $u_h \in V_h^{+,iu}$ is an admissible function by construction.

Proof of Lemma 4.1. In addition, we know that at the minimum it holds

$$\langle J'(R_h u)(v_h - R_h u) \rangle \geq 0$$

for all admissible $v_h \in V_h^+ \cap (iu)^\perp$. We compute that

$$\begin{aligned} J(w_h + v_h) - J(w_h) &= \frac{1}{2} a_\kappa(u - w_h - v_h, u - w_h - v_h) - \frac{1}{2} a_\kappa(u - w_h, u - w_h) \\ &= \frac{1}{2} a_\kappa(v_h, v_h) - a_\kappa(u - w_h, v_h) \end{aligned}$$

such that

$$\langle J'(w_h)v_h \rangle = a_\kappa(w_h - u, v_h)$$

and hence, it holds

$$0 \leq \langle J'(R_h u)(v_h - R_h u) \rangle = a_\kappa(R_h u - u, v_h - R_h u)$$

which is equivalent to the assertion. $\square$

4.2. Derivation of the critical point equation. We express the minimization with the help of the functions

$$g_i(v_h) = \frac{1}{2}(|v_h(x_i)|^2 - 1), \quad i = 1, \dots, N,$$

equivalently as

$$\min_{v_h \in V_h} E(v_h) \quad \text{subject to} \quad g_i(v_h) \leq 0, \quad i = 1, \dots, N$$

Then, we obtain the Lagrangian function

$$F(v_h, \nu) = E(v_h) + \nu \cdot g$$

Now, let $u_h \in V_h^+$ be some minimizer of (3.2), then there exists some μ such that (u_h, μ) satisfy for all $\varphi_h \in V_h$

$$(4.2) \quad \begin{aligned} 0 &= E'(u_h)\varphi_h + \sum_{i=1}^N \mu_i \operatorname{Re}(u_h(x_i)\varphi_h(x_i)^*) \\ g(u_h) &\leq 0, \quad \mu \geq 0, \quad \mu \cdot g(u_h) = 0. \end{aligned}$$

4.3. Discrete error. We follow the strategy of [9, Section 5], and adapt the arguments to the constraint case. We decompose the error in its best-approximation part and the discrete part via

$$\|u - u_h\|_{H_\kappa^1} \leq \|u - R_h u\|_{H_\kappa^1} + \|R_h u - u_h\|_{H_\kappa^1}$$

The first part is part of the estimate in the theorem, and for the second part, we have by the coercivity of $\partial_u^2 E(u)$ in Lemma 2.1

$$\begin{aligned} \|R_h u - u_h\|_{H_\kappa^1}^2 &\leq \rho_\kappa(u) \partial_u^2 E(u)[R_h u - u_h, R_h u - u_h] \\ &= \rho_\kappa(u) \partial_u^2 E(u)[R_h u - u, R_h u - u_h] + \rho_\kappa(u) \partial_u^2 E(u)[u - u_h, R_h u - u_h] \\ &= \rho_\kappa(u) \varepsilon_h^{\text{lin}}(R_h u - u_h) + \rho_\kappa(u) \varepsilon_h^{\text{nonlin}}(R_h u - u_h) \end{aligned}$$

With this we can proof our first main result.

Proof of Theorem 3.1. We are only left to estimate the linear and nonlinear part. For the linear part, we compute

$$\begin{aligned} &\varepsilon_h^{\text{lin}}(R_h u - u_h) \\ &= a_\kappa(R_h u - u, R_h u - u_h) + \beta(R_h u - u, R_h u - u_h)_{L^2} \\ &\quad + \operatorname{Re} \int_\Omega (2|u|^2 - 1)(R_h u - u)(R_h u - u_h)^* + u^2(R_h u - u)^*(R_h u - u_h)^* \, dx \\ &\leq a_\kappa(R_h u - u, R_h u - u_h) + (2 + \beta)\|R_h u - u\|_{L^2}\|R_h u - u_h\|_{L^2}, \\ &\leq (2 + \beta)\|R_h u - u\|_{L^2}\|R_h u - u_h\|_{L^2}, \end{aligned}$$

where we used the property (4.1) for the last inequality. For the nonlinear part we employ the relation (2.1) to obtain

$$\begin{aligned}
\varepsilon_h^{\text{nonlin}}(R_h u - u_h) &= \partial_u^2 E(u)[u - u_h, R_h u - u_h] \\
&= \partial_u^2 E(u)[u, R_h u - u_h] - \partial_u^2 E(u_h)[u_h, R_h u - u_h] + \partial_u^2 E(u_h)[u_h, R_h u - u_h] - \partial_u^2 E(u)[u_h, R_h u - u_h] \\
&= \operatorname{Re} \int_{\Omega} 2|u|^2 u (R_h u - u_h)^* \, dx - \langle \partial_u E(u_h)(R_h u - u_h) \rangle - \operatorname{Re} \int_{\Omega} 2|u_h|^2 u_h (R_h u - u_h)^* \, dx \\
&\quad + \operatorname{Re} \int_{\Omega} (2|u_h|^2 - 1) u_h (R_h u - u_h)^* + u_h^2 u_h^* (R_h u - u_h)^* \, dx \\
&\quad - \operatorname{Re} \int_{\Omega} (2|u|^2 - 1) u_h (R_h u - u_h)^* + u^2 u_h^* (R_h u - u_h)^* \, dx \\
&= \operatorname{Re} \int_{\Omega} (2|u|^2 u - 2|u_h|^2 u_h + (2|u_h|^2 - 1)u_h + u_h^2 u_h^* - (2|u|^2 - 1)u_h + u^2 u_h^*) (R_h u - u_h)^* \, dx \\
&\quad - \langle \partial_u E(u_h)(R_h u - u_h) \rangle \\
&= \operatorname{Re} \int_{\Omega} (2u|u - u_h|^2 + (u - u_h)^2 u^* - |u - u_h|^2 (u - u_h)) (R_h u - u_h)^* \, dx - \langle \partial_u E(u_h)(R_h u - u_h) \rangle.
\end{aligned}$$

Note that in the unconstrained case the last term would vanish. Here, we obtain from the critical point equation (4.2) that

$$\begin{aligned}
-\langle \partial_u E(u_h)(R_h u - u_h) \rangle &= \sum_{i=1}^M \mu_i \operatorname{Re}(u_h(x_i)(R_h u - u_h)(x_i)^*) \\
&= \sum_{i=1}^M \mu_i (\operatorname{Re}(u_h(x_i) R_h u(x_i)^*) - |u_h(x_i)|^2) \\
&\leq \sum_{i=1}^M \mu_i (|u_h(x_i)| |R_h u(x_i)| - |u_h(x_i)|^2) = (1 + \delta) \sum_{i \in A} \mu_i (|R_h u(x_i)| - (1 + \delta)) \leq 0
\end{aligned}$$

since by construction of the Ritz projection $|R_h u(x_i)| \leq 1 + \delta$ at all nodes, and where A contains the active set $\mu_i > 0$ where $|u_h(x_i)| = 1 + \delta$. Thus, we can conclude

$$\begin{aligned}
\varepsilon_h^{\text{nonlin}}(R_h u - u_h) &\leq \operatorname{Re} \int_{\Omega} (2u|u - u_h|^2 + (u - u_h)^2 u^* - |u - u_h|^2 (u - u_h)) (R_h u - u_h)^* \, dx \\
&\leq (3\|u - u_h\|_{L^4}^2 + \|u - u_h\|_{L^6}^3) \|R_h u - u_h\|_{L^2}.
\end{aligned}$$

Adding both estimates gives the desired estimate for part (a). Absorption directly gives part (b) for h sufficiently small. $\square$

5. APPROXIMATION OF THE CONSTRAINED RITZ PROJECTION IN H_{κ}^1 AND EQUIVALENT FORMULATION

In this section, we present the approximation property in the natural norm H_{κ}^1 by constructing a suitable upper bound, and second provide the framework for the L^2 estimate.

5.1. Approximation in H_{κ}^1 . Here, we give the proof of the H_{κ}^1 bound for R_h . We use a suitable projection combined with a phase shift.

Proof of Theorem 3.2. Let N_j be the patch around the node x_j , and define operator

$$\tilde{\pi}: \{w \in L^{\infty}(\Omega) \mid |w| \leq 1 \text{ a.e.}\} \rightarrow V_h^+, \quad (\tilde{\pi}w)(x_j) = \frac{1}{|N_j|} \int_{N_j} w(x) \, dx$$

which is H^1 -stable and has optimal approximation properties in L^p .

(1) Next, we define the angle ψ_h such that $e^{i\psi_h}\tilde{\pi}u \perp iu$ and obtain **B: checken nochmal**

$$|\psi_h| \lesssim \frac{|(\tilde{\pi}u, iu)|}{|(\tilde{\pi}u, u)|}$$

and thus

$$\begin{aligned} \|u - e^{i\psi_h}\tilde{\pi}u\|_{H_\kappa^1} &\leq \|u - \tilde{\pi}u\|_{H_\kappa^1} + \|(1 - e^{i\psi_h})\tilde{\pi}u\|_{H_\kappa^1} \\ &\lesssim \|u - \tilde{\pi}u\|_{H_\kappa^1} + \frac{|(\tilde{\pi}u, iu)|}{|(\tilde{\pi}u, u)|} \|\tilde{\pi}u\|_{H_\kappa^1} \end{aligned}$$

(2) We are left to estimate the fraction.

$$\frac{|(\tilde{\pi}u, iu)|}{|(\tilde{\pi}u, u)|} \|\tilde{\pi}u\|_{H_\kappa^1} \lesssim \frac{\|\tilde{\pi}u - u\|_{L^2} \|u\|_{L^2}}{|(\tilde{\pi}u - u, u) + \|u\|_{L^2}^2|} \|\tilde{\pi}u\|_{H_\kappa^1}$$

and with $\|u\|_{H_\kappa^1} \lesssim \|u\|_{L^2}$

$$\|\tilde{\pi}u - u\|_{L^2} \lesssim h\kappa \|u\|_{H_\kappa^1} \lesssim h\kappa \|u\|_{L^2}$$

we obtain for $\kappa h \lesssim 1$

$$\frac{\|\tilde{\pi}u - u\|_{L^2}}{|(\tilde{\pi}u - u, u) + \|u\|_{L^2}^2|} \|\tilde{\pi}u\|_{H_\kappa^1} \lesssim \frac{\|\tilde{\pi}u - u\|_{L^2} \|u\|_{L^2}}{\|u\|_{L^2}^2 - h\kappa \|u\|_{L^2}^2} \|u\|_{L^2}^2 \lesssim \|\tilde{\pi}u - u\|_{L^2}$$

□

5.2. Check equivalence with constrained bilinear forms. We now aim at an equivalent formulation of the Ritz projection via a stabilized nonlinear variational formulation introduced in [4]. As this was done for real-valued problem, we investigate the necessary modifications. We define the bilinear form and norm

$$s(v_h, w_h) = C_{\text{stab}} \sum_i^N \alpha_h(x_i) \operatorname{Re} v_h(x_i) w_h(x_i)^*, \quad \|v_h\|_s^2 = s(v_h, v_h),$$

with some to be chosen positive function α_h . With the definition of V_h^+ and $V_h^{+,iu}$ in (3.1), we can check the following properties, cf. [4, Lemma 3.1]

Lemma 5.1. *The bilinear form s satisfies*

$$\begin{aligned} s(v_h^- - w_h^-, v_h^+ - w_h^+) &\geq 0, \quad \text{for all } v_h, w_h \in V_h \\ s(v_h^-, w_h - v_h^+) &\leq 0, \quad \text{for all } v_h \in V_h, w_h \in V_h^+ \end{aligned}$$

Proof. We only have to proof the first inequality, as it implies the second. Note that it is sufficient to check this for every nodal point x_i , since the weight α_h is positive, and we thus fix some x_i . We have to distinguish three cases.

(1) If $|v_h(x_i)|, |w_h(x_i)| \leq 1 + \delta$ then the negative parts both vanish, and the estimate holds.

(2) If $|v_h(x_i)|, |w_h(x_i)| > 1 + \delta$, we have

$$\begin{aligned} &= \operatorname{Re} \left(v_h(x_i) - v_h(x_i) \frac{1+\delta}{|v_h(x_i)|} - w_h(x_i) + w_h(x_i) \frac{1+\delta}{|w_h(x_i)|} \right) (1+\delta) \left(\frac{v_h(x_i)}{|v_h(x_i)|} - \frac{w_h(x_i)}{|w_h(x_i)|} \right)^* \\ &= -(1+\delta)^2 \left| \frac{v_h(x_i)}{|v_h(x_i)|} - \frac{w_h(x_i)}{|w_h(x_i)|} \right|^2 + (1+\delta) \operatorname{Re} \left(v_h(x_i) - w_h(x_i) \right) \left(\frac{v_h(x_i)}{|v_h(x_i)|} - \frac{w_h(x_i)}{|w_h(x_i)|} \right)^* \geq 0 \end{aligned}$$

using the first part of Lemma A.1.

(3) If $|v_h(x_i)| > 1 + \delta$ and $|w_h(x_i)| \leq 1 + \delta$ (which is enough by symmetry), we have $w_h^-(x_i) = 0$ and hence

$$\begin{aligned} &\operatorname{Re} \left(v_h(x_i) - (1+\delta) \frac{v_h(x_i)}{|v_h(x_i)|} \right) \left((1+\delta) \frac{v_h(x_i)}{|v_h(x_i)|} - w_h(x_i) \right)^* \\ &= - \left| (1+\delta) \frac{v_h(x_i)}{|v_h(x_i)|} - w_h(x_i) \right|^2 + \operatorname{Re} \left(v_h(x_i) - w_h(x_i) \right) \left((1+\delta) \frac{v_h(x_i)}{|v_h(x_i)|} - w_h(x_i) \right)^* \geq 0 \end{aligned}$$

using the second part of Lemma A.1. □

Further, we have to show that the a_κ -norm can be controlled with the s -bilinear form on the space V_h . This is given by the following result, cf eq. (3.8) in [4].

Lemma 5.2. *Define*

$$\alpha_h(x_i) = h(x_i)^d (\kappa^{-2} h(x_i)^{-2} + \|A\|_{L^\infty} \kappa^{-1} h(x_i)^{-1} + \|A\|_{L^\infty}^2)$$

then there is a constant C such that

$$\|v_h\|_{a_\kappa}^2 \leq \frac{C}{C_{stab}} \|v_h\|_s^2$$

where C only depends on the shape-regularity of the mesh.

Proof. We denote by Ω_i the union of all elements touching x_i . Then

$$\begin{aligned} & \operatorname{Re} \int_{\Omega_i} \left| \frac{i}{\kappa} \nabla v_h + \mathbf{A} v_h \right|^2 + \beta |v_h|^2 \, dx \\ &= \operatorname{Re} \int_{\Omega_i} \left| \frac{1}{\kappa} \nabla v_h \right|^2 + (|\mathbf{A}|^2 + \beta) |\mathbf{A} v_h|^2 \, dx + \operatorname{Re} \int_{\Omega_i} \left(\frac{i}{\kappa} \nabla v_h \right) \cdot (\mathbf{A} v_h)^* \\ &\lesssim (\kappa^{-2} h(x_i)^{-2} + (\|A\|_{L^\infty}^2 + \beta) + \|A\|_{L^\infty} \kappa^{-1} h(x_i)^{-1}) \|v_h\|_{L^2(\Omega_i)}^2 \end{aligned}$$

with a constant C that only depends on the shape regularity. Using the equivalence to the mass lumped inner product and then summing up over all Ω_i gives the claim. $\square$

This allows us to define the following bilinear form

$$(5.1) \quad a_h(u_h; v_h) = a_\kappa(u_h^+, v_h) + s(u_h^-, v_h)$$

and with this the (nonlinear) problem: Find $u_h \in V_h^{iu}$ such that

$$(5.2) \quad a_h(u_h; v_h) = a_\kappa(u, v_h), \quad \text{for all } v_h \in V_h^{iu}.$$

Theorem 5.3. *The problem (5.2) has a unique solution $u_h \in V_h^{iu}$, and u_h^+ coincides with the minimizer of (3.3), i.e., $u_h^+ = R_h u$.*

Proof. We may proceed along the lines of [4, Theorem 3.2] to obtain a unique solution of (5.2). By [4, Theorem 3.5], we can derive that u_h^+ satisfies the variational inequality of Lemma 4.1, and by uniqueness we obtain the equivalence of the solutions. $\square$

Let us emphasize that at no point, we require the computation of R_h neither by minimization nor by the variational inequality nor the nonlinear equation. it solely serves as a theoretical tool in the analysis.

6. APPROXIMATION IN L^2 , THE CASE $\delta > 0$

In this section, we provide the proof of Theorem 3.3. The main idea is to use the relaxation by $\delta > 0$ to set up a bootstrapping argument. We first show that the Ritz projection is bounded in L^∞ by something strictly smaller than $1 + \delta$, which means not there is no active node for the constraint.

Lemma 6.1. *Let $u \in H^2(\Omega)$ be a minimizer of (1.1).*

(a) *If $d = 2$, then for $C_\infty \kappa^2 |\log h| h < \delta$, it holds*

$$\|R_h u\|_{L^\infty} < 1 + \delta.$$

(a) *If $d = 3$, then for $C'_\infty \kappa^2 h^{1/2} < \delta$, it holds*

$$\|R_h u\|_{L^\infty} < 1 + \delta.$$

Note that this constraint is in two dimensions comparable to the resolution condition in [6, Theorem 3.2], whereas there even appears the $\rho_\kappa(u)$ as a factor. If $\rho_\kappa(u)$ would grow at least like $\kappa^{3/2}$, then the resolution condition in the three dimensional case implies our condition.

Proof of Lemma 6.1. Let for the moment $\widehat{u}_h \in V_h$ be arbitrary. Then, we estimate **B: with the estimate** $\|v\|_{L^p} \lesssim \sqrt{p}\|v\|_{H^1}$, quote needed

$$\begin{aligned} \|u - R_h u\|_{L^\infty} &\leq \|u - \widehat{u}_h\|_{L^\infty} + \|\widehat{u}_h - R_h u\|_{L^\infty} \\ &\lesssim \|u - \widehat{u}_h\|_{L^\infty} + h^{-d/p} \|\widehat{u}_h - R_h u\|_{L^p}, \end{aligned}$$

and we will employ the different Sobolev embedding for the different spatial dimensions.

(a) For $d = 2$, we have

$$\begin{aligned} h^{-d/p} \|\widehat{u}_h - R_h u\|_{L^p} &\leq h^{-2/p} p \|\widehat{u}_h - R_h u\|_{H^1} \\ &\lesssim h^{-2/p} \sqrt{p} \|\widehat{u}_h - u\|_{H^1} + h^{-2/p} \sqrt{p} \kappa \|u - R_h u\|_{H^1_\kappa} \end{aligned}$$

Choosing p such that $h^{-2/p} \sqrt{p} \sim |\log h|^{1/2}$, and $\widehat{u}_h = I_h u$, the estimate from Theorem 3.2, and the fact that $\kappa^{-2} \|u\|_{H^2} \lesssim 1$, we obtain

$$\|u - R_h u\|_{L^\infty} \lesssim h \|u\|_{H^2} + |\log h| (h \|u\|_{H^2} + \kappa h) \leq C_\infty \kappa^2 |\log h| h$$

and in particular for $C_\infty \kappa^2 |\log h| h^s < \delta$, it holds

$$\|R_h u\|_{L^\infty} \leq \|u\|_{L^\infty} + \|u - R_h u\|_{L^\infty} < 1 + \delta,$$

which gives the claim.

(b) For $d = 3$, we choose $p = 6$ to obtain

$$h^{-3/6} \|\widehat{u}_h - R_h u\|_{L^6} \lesssim h^{-1/2} \|\widehat{u}_h - u\|_{H^1} + h^{-1/2} \kappa \|u - R_h u\|_{H^1_\kappa} \lesssim h^{1/2} \kappa^2$$

and thus with some $C'_\infty > 0$, we obtain for $C'_\infty \kappa^2 h^{1/2} < \delta$ the bound

$$\|R_h u\|_{L^\infty} \leq \|u\|_{L^\infty} + \|u - R_h u\|_{L^\infty} < 1 + \delta,$$

which gives the second claim. $\square$

Now looking at the formulation (5.1), we see that the corrected part of $R_h u$ vanishes, and in fact satisfies instead of (5.2) the linear version

$$a_\kappa(R_h u, v_h) = a_\kappa(u, v_h), \quad \text{for all } v_h \in V_h^{iu}.$$

In particular, we recover the orthogonality $a_\kappa(R_h u - u, v_h) = 0$ for all $v_h \in V_h^{iu}$.

Proof of Theorem 3.3. We follow the standard Aubin-Nitsche trick. Let $w \in V_h^{iu}$ be the unique solution to

$$a_\kappa(R_h u, \varphi) = (R_h u - u, \varphi)$$

for all $\varphi \in H_{iu}^1$. Hence, we obtain since $R_h u - u \in H_{iu}^1$ and by the orthogonality for all $w_h \in V_h^{iu}$

$$\|R_h u - u\|^2 = a_\kappa(w, R_h u - u) \leq \|w - w_h\|_{a_\kappa} \|R_h u - u\|_{a_\kappa}$$

the regularity results and best approximation in [9], then give the two claimed estimates. $\square$

7. APPROXIMATION IN L^2 , THE CASE $\delta = 0$

In this section, we provide the proof of Theorem 3.4. The main effort consists of translating the positive constraint into the modulus constraint. However, we have to make a modification in the orthogonality constraint, i.e. instead of looking for a discrete minimizer in a subset of $(iu)^\perp$, we use a subspace of $(I_h iu)^\perp$, assuming that $I_h u \neq 0$, and comment on the necessary changes in Section 7.1. We therefore, study the problem as in [3] in the form

$$a(\phi, v) = (\mu \phi, v) + (\beta \phi', v) + \gamma(\phi', v'), \quad \text{with } (\phi, v) = \operatorname{Re} \int_\Omega \phi v^* \, dx$$

with possibly space dependent μ, β, γ . We later have the scaling

$$\mu \sim 1, \quad \beta \sim \kappa^{-1}, \quad \gamma \sim \kappa^{-2}$$

We also assume that there is a constant such that

$$\|\phi\|_\mu \leq C \|\phi\|_a$$

Let on we replace a by a_κ , but for the moment we study

$$a(u, v) = f(v), \quad \text{for all } v \in V_0 = \{v \in H^1(\Omega) \mid (I_h v, I_h i u) = 0\},$$

and we assume for the exact solution

$$|u(x)| \leq 1 \text{ almost everywhere in } \Omega.$$

The bilinear form s is an adaption of the above construction with $\kappa = 1$ and $A = \text{const.}$

Choice of the weights: We define the complex weights

$$\omega_j = -\frac{u_h(x_j)^-}{u(x_j) - u_h(x_j)^+}$$

with the convention $0/0 = 0$, and observe the following.

Lemma 7.1. *Either, we have $u(x_j) = u_h(x_j)^+$ or the weights satisfy*

$$\operatorname{Re} \omega_j \geq 0$$

for all j .

Proof. If $|u_h(x_j)| \leq 1$ the weight is zero. Hence, we only have to consider $|u_h(x_j)| > 1$. Thus, by positivity of the absolute value and the minus sign in front of the weights, we obtain

$$\begin{aligned} 0 \leq \operatorname{Re} \omega_j &\iff 0 \leq \operatorname{Re}(u_h(x_j)^-(u(x_j) - u_h(x_j)^+)^*) \\ &= \operatorname{Re}\left((u_h(x_j) - \frac{u_h(x_j)}{|u_h(x_j)|})(u(x_j) - \frac{u_h(x_j)}{|u_h(x_j)|})^*\right) \\ &= -\left|(u_h(x_j) - \frac{u_h(x_j)}{|u_h(x_j)|})\right|^2 + \operatorname{Re}\left((u_h(x_j) - \frac{u_h(x_j)}{|u_h(x_j)|})(u(x_j) - u_h(x_j))^*\right) \end{aligned}$$

and it is sufficient to show that the second part is non-positive. We denote by $c = \operatorname{Re} u_h(x_j)u(x_j)^*$ with $|c| \leq |u_h(x_j)|$, and neglect the point evaluation in the following:

$$\begin{aligned} \operatorname{Re}\left((u_h - \frac{u_h}{|u_h|})(u - u_h)^*\right) &= c - \frac{c}{|u_h|} - |u_h|^2 + |u_h| \\ &= \frac{c}{|u_h|^2}(|u_h|^2 - |u_h|) - |u_h|^2 + |u_h| \\ &= \left(\frac{c}{|u_h|^2} - 1\right)(|u_h|^2 - |u_h|) \leq 0 \end{aligned}$$

since

$$\frac{c}{|u_h|^2} - 1 \leq 0 \quad \text{and} \quad |u_h|^2 - |u_h| \geq 0,$$

which gives the claim. $\square$

We define

$$\begin{aligned} J_0 &= \{x_j \mid u_h^+(x_j) = u(x_j) \text{ and } u_h^-(x_j) \neq 0\} \\ \mathcal{H}_0 &= \{v \in V_0 \mid \operatorname{Re}(u_h(x_j)v(x_j)^*) = 0, \text{ for all } j \in J_0\} \end{aligned}$$

and note that $\omega_j = \infty$ if and only if $j \in J_0$, but in the following these weights will only be multiplied by zero. Further, we mention that for $j \in J_0$ the difference $u(x_j) - u_h^+(x_j) = 0$ and thus the error $u(x_j) - u_h^+(x_j) \in \mathcal{H}_0$ is a valid test function later on.

With the weights, we define for $\phi, v \in \mathcal{H}_0$ the stabilization

$$\tilde{s}(\phi, v) = C_{\text{stab}} \sum_i^N \alpha_h(x_i) \operatorname{Re}(\omega_i \phi(x_i) v(x_i)^*),$$

which directly gives by Lemma 7.1

$$\tilde{s}(\phi, \phi) = C_{\text{stab}} \sum_i^N \alpha_h(x_i) \operatorname{Re} \omega_i |\phi(x_i)|^2 \geq 0.$$

For the duality argument, we require the identity

Lemma 7.2. *It holds the equality*

$$\widetilde{s}(u - u_h^+, v) = -s(u_h^-, v),$$

for all $v \in \mathcal{H}_0$.

Proof. We compute the sum and note the whenever $i \notin J_0$ it holds $\omega_i(u(x_i) - u_h^+(x_i)) = -u_n(x_i)^-$ and thus

$$\begin{aligned} \widetilde{s}(u - u_h^+, v) + s(u_h^-, v) &= C_{\text{stab}} \sum_{i=1}^N \alpha_h(x_i) \operatorname{Re} \left(\omega_i(u(x_i) - u_h^+(x_i)) + u_h^-(x_i) \right) v(x_i)^* \\ &= C_{\text{stab}} \sum_{i \in J_0}^N \alpha_h(x_i) \operatorname{Re} u_h^-(x_i) v(x_i)^* \\ &= C_{\text{stab}} \sum_{i \in J_0}^N \alpha_h(x_i) \operatorname{Re}(u_h(x_i) v(x_i)^*) (1 - |u_h(x_i)|^{-1}) = 0 \end{aligned}$$

by the definition of $\mathcal{H}_0$. $\square$

Lemma 7.3. *Let $\vartheta \in \mathcal{H}_0$ and $\omega_j \in \mathbb{C}$ some weights. Then, there exists $\vartheta_h \in V_h \cap \mathcal{H}_0$ such that*

$$\gamma(w', \vartheta'_h) = -C_{\text{stab}} \sum_i^N \alpha_h(x_i) \operatorname{Re}(\omega_i w(x_i) \vartheta(x_i)^*), \text{ for all } w \in \mathcal{H}_0.$$

Proof. Since we have Neumann conditions, we eliminate the constants to obtain uniqueness. We again proceed as in [3, Lemma 3.3], and let ϑ_h be the solution, when we replace $w \in \mathcal{H}_0$ by $w \in \mathcal{H}_0 \cap V_h$. In particular, we may insert $I_h w$ for all $w \in \mathcal{H}_0$ and only have to show

$$\operatorname{Re}(w' - (I_h w)', \vartheta'_h) = 0$$

however, on each sub interval using integration by parts, and the fact that ϑ'_h is constant on the sub interval

$$\operatorname{Re} \int_{x_i}^{x_{i+1}} (w' - (I_h w)') \vartheta'_h \, dx = [(w - I_h w) \vartheta'_h]_{x_i}^{x_{i+1}} = 0$$

as the interpolation $w(x_j) - I_h w(x_j) = 0$. $\square$

We consider the adjoint problem for $\psi \in L^2$ find $\vartheta \in \mathcal{H}_0$ such that

$$\widetilde{a}(w, \vartheta) + \widetilde{s}(w, \vartheta) = (\mu w, \psi)$$

for all $w \in \mathcal{H}_0$, where $\widetilde{a}$ is the same as a and the derivative of the mixed term is on the other side.

Lemma 7.4 (Lemma 3.2 in [3]). *There is a unique solution $\vartheta \in \mathcal{H}_0$ and satisfies*

$$\|\vartheta\|_\mu + \|\vartheta\|_{\widetilde{a}} \lesssim \|\psi\|_\mu$$

with constants independent of the coefficients.

Proof. Existence and uniqueness are clear, and we only insert

$$\|\vartheta\|_\mu \lesssim \widetilde{a}(\vartheta, \vartheta) \leq \widetilde{a}(\vartheta, \vartheta) + \widetilde{s}(\vartheta, \vartheta) = (\mu \vartheta, \psi) \leq \|\vartheta\|_\mu \|\psi\|_\mu$$

and hence the claims. $\square$

Lemma 7.5 (Lemma 3.3 in [3]). *Consider $\psi \in L^2$ and let $\vartheta \in \mathcal{H}_0$ solve the adjoint problem. Then, the $\vartheta_2 \in H^2(\mathcal{T}_h)$, and satisfies*

$$\gamma \|\vartheta_2\|_{H^{3/2-\varepsilon}(\mathcal{T}_h)} \lesssim (\mu^{1/2} + \beta) \|\psi\|_\mu$$

Proof. We define $\vartheta_2 = \vartheta - \vartheta_h$. Then, we observe that

$$\begin{aligned}\gamma(w', \vartheta'_2) &= -(\mu w, \vartheta) - (\beta w, \vartheta') - \tilde{s}(w, \vartheta) + (\mu w, \psi) + C_{\text{stab}} \sum_i^N \alpha_h(x_i) \operatorname{Re}(\omega_i w(x_i) \vartheta(x_i)^*) \\ &= -(w, \mu \vartheta + \beta \vartheta' + \mu \psi)\end{aligned}$$

We denote by $f = \mu \vartheta + \beta \vartheta' + \mu \psi \in L^2$, and show regularity of ϑ_2 which satisfies

$$\gamma(w', \vartheta'_2) = (w, f) \text{ for all } w \in \mathcal{H}_0.$$

To obtain H^2 , we write on each subinterval I^ℓ : $w = \widehat{w} + \alpha(I_h i u) \in H^1(I^\ell)$ with

$$\alpha = \frac{(I_h w, I_h i u)}{\|I_h u\|^2}, \quad |\alpha| \leq \frac{\|I_h w\|}{\|I_h u\|} \lesssim \frac{\|w\|_{H^{1/2+\varepsilon}}}{\|I_h u\|}$$

with $\widehat{w} \in \mathcal{H}_0$. In addition, using the bounds on u from Lemma 2.2, we conclude further for $\kappa h \lesssim 1$

$$\|I_h u\| \geq (1 - c\kappa h)\|u\| \geq C(1 - c\kappa h)\|u\|_{H_\kappa^1}.$$

Then for all $w \in H^1(I^\ell)$

$$\gamma(w', \vartheta'_2) = \gamma(\widehat{w}', \vartheta'_2) + \gamma\alpha(iu', \vartheta'_2) = (\widehat{w}, f) + \gamma\alpha(iu', \vartheta'_2) = (w, f) - \alpha(iu, f) + \gamma\alpha(iu', \vartheta'_2)$$

where one can show is in Lemma 2.8 in [9] that $|\gamma\alpha(iu', \vartheta'_2)| \lesssim \|\alpha iu\|_{L^2} \|f\|_{L^2}$ and by orthogonality $\|\alpha iu\|_{L^2} \leq \|w\|_{L^2}$ such that

$$\gamma(w', \vartheta'_2) = (w, \widetilde{f}) \text{ for all } w \in H^1(I_\ell)$$

with uniformly bounded $\widetilde{f} \in H^{-1/2-\varepsilon}$, and the local $H^{3/2-\varepsilon}$ -regularity is restored. $\square$

We can now conclude along the lines of Theorem 3.4 in [3].

Theorem 7.6. *Under the assumptions of the preceding results, we obtain the bound*

$$\begin{aligned}\|u - R_h u\|_\mu &\lesssim \left(\mu^{1/2} \gamma^{-1} h^{3/2-\varepsilon} \|u - R_h u\|_\mu + \beta \|u' - (R_h u)'\| h^{1/2-\varepsilon} \right. \\ &\quad \left. + \gamma^{-1/2} h^{1/2-\varepsilon} \gamma^{1/2} \|u' - (R_h u)'\| \right) (\mu^{1/2} + \beta)\end{aligned}$$

with constants independent of the coefficients and h .

Inserting now the values of the coefficients, we arrive at

$$\|u - R_h u\| \lesssim \kappa^2 h^{3/2-\varepsilon} \|u - R_h u\| + \kappa h^{1/2-\varepsilon} \|u - R_h u\|_a$$

and for $\kappa^2 h^{3/2-\varepsilon}$ sufficiently small, we have shown the claim of Theorem 3.4.

Proof of Theorem 7.6. Let $\xi \in \mathcal{H}_0$ be the solution for $\psi = u - R_h u$. Then, we derive along the lines of [3]

$$\|u - R_h u\|_\mu^2 = a(u - R_h u, \xi - I_h \xi)$$

where we used that also $I_h \xi \in \mathcal{H}_0$ and conclude

$$\begin{aligned}\|u - R_h u\|_\mu^2 &\lesssim \left(\mu^{1/2} \gamma^{-1} h^{3/2-\varepsilon} \|u - R_h u\|_\mu + \beta \|u' - (R_h u)'\| h^{1/2-\varepsilon} \right. \\ &\quad \left. + \gamma^{-1/2} h^{1/2-\varepsilon} \gamma^{1/2} \|u' - (R_h u)'\| \right) \gamma \|\vartheta_2\|_{H^{3/2-\varepsilon}} \\ &\lesssim \left(\mu^{1/2} \gamma^{-1} h^{3/2-\varepsilon} \|u - R_h u\|_\mu + \beta \|u' - (R_h u)'\| h^{1/2-\varepsilon} \right. \\ &\quad \left. + \gamma^{-1/2} h^{1/2-\varepsilon} \gamma^{1/2} \|u' - (R_h u)'\| \right) (\mu^{1/2} + \beta) \|u - R_h u\|_\mu\end{aligned}$$

and hence the claim. $\square$

7.1. Necessary modification for the new orthogonality constraint. The main modification is the coercivity of E'' which only helps perpendicular to iu . However, let $\varphi_h \perp I_h iu$. We decompose this as

$$\varphi_h = \widehat{\varphi} + \alpha iu, \quad \widehat{\varphi} \perp iu, \quad \alpha = \frac{(\varphi_h, iu)}{\|u\|^2}.$$

Thus, we compute

$$\|\varphi_h\|_{H_\kappa^1}^2 \lesssim \|\widehat{\varphi}\|_{H_\kappa^1}^2 + \|\alpha iu\|_{H_\kappa^1}^2 \lesssim \rho_\kappa(u) E''(\widehat{\varphi}, \widehat{\varphi}) + \frac{|(\varphi_h, iu)|^2}{\|u\|_{H_\kappa^1}^2}$$

where we used the coercivity and Lemma 2.2 for the norm exchange for u . Further, we obtain

$$|(\varphi_h, iu)| = |(\varphi_h, iu - I_h iu)| \lesssim \|\varphi_h\|_{H_\kappa^1} \|\kappa h\|_{H_\kappa^1} \|u\|_{H_\kappa^1}$$

and we can absorb it to the left hand side for κh small. Further, since $E''(\widehat{\varphi}, \widehat{\varphi}) = E''(\varphi_h, \varphi_h)$ by the kernel property, we obtain coercivity.

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APPENDIX A. USEFUL ESTIMATES

Lemma A.1. *Let $u, v \in \mathbb{C}$, and $\delta \geq 0$.*

(a) *For $|u|, |v| \geq 1 + \delta$ it holds*

$$\operatorname{Re}\left(\frac{u}{|u|} - \frac{v}{|v|}\right)(u - v)^* \geq (1 + \delta) \left| \frac{u}{|u|} - \frac{v}{|v|} \right|^2$$

(b) *For $|u| \geq 1 + \delta$ and $|v| \leq 1 + \delta$ it holds*

$$\operatorname{Re}\left((1 + \delta)\frac{u}{|u|} - v\right)(u - v)^* \geq \left|(1 + \delta)\frac{u}{|u|} - v\right|^2$$

Proof. (a) We write $u = ra$, $v = sb$ for $|a| = |b| = 1$ and denote by $c = \operatorname{Re} ab^*$. With this, we compute $|a - b|^2 = 2(1 - c)$ and further

$$\operatorname{Re}\left((a - b)(ra - sb)^*\right) = r(1 - c) - s(c - 1) = (r + s)(1 - c) = \frac{r + s}{2}|a - b|^2.$$

Noting that $r + s = |a| + |b| \geq 2(1 + \delta)$, gives the claim.

(b) Again, we write $u = ra$ for $|a| = 1$ and set $c = \operatorname{Re} ab^*$.

$$|(1 + \delta)a - b|^2 = (1 + \delta)^2 - 2c(1 + \delta) + |b|^2$$

and further

$$\operatorname{Re}\left(((1 + \delta)a - b)(ra - b)^*\right) = r(1 + \delta - c) - c(1 + \delta) + |b|^2$$

Since $|c| \leq 1 + \delta$ and $r \geq 1 + \delta$, this implies

$$\operatorname{Re}\left((a - b)(ra - b)^*\right) \geq (1 + \delta)(1 + \delta - c) - c(1 + \delta) + |b|^2 = |a - b|^2,$$

and hence the claim. □

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